Social & Media Platform APIs

Week 12 · APIs module · Textbook Chapter 12

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This week at a glance

Last week the data came from governments obligated to be open. This week it comes from private platforms that share it *at their discretion* — and can stop.

Monday | Concepts & notebook intro

- OAuth2, four platforms, and the governance spectrum from proprietary to open

Wednesday | Notebook lab

- Reddit, Spotify, Bluesky, and Mastodon — authenticate, retrieve, analyze

Friday | Textbook revisions

- Propose & peer-review pull requests improving Chapter 12

Companion notebook

`ch-12-social.ipynb`

Bring a laptop

Wednesday is hands-on — register your own apps and pull live data.

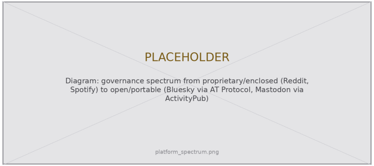
1. Monday · Concepts

Platforms, data, and power

Social platforms mediate an enormous share of public discourse. Their APIs are a **different animal** from last week's government APIs (Ch. 11): those exist because of transparency mandates; these exist at the discretion of a private company.

So access can change overnight — Reddit's 2023 pricing shift and Twitter's post-acquisition lockout are the cautionary cases.

We study four platforms spanning the spectrum from **proprietary** (Reddit, Spotify) to **open** (Bluesky, Mastodon) — the enclosure-vs-openness tension from the post-API age (Ch. 2).



Proprietary to open: who controls access.

Textbook Ch. 12, "Platforms, Data, and Power." Cross-refs: Government APIs (Ch. 11), the post-API age (Ch. 2).

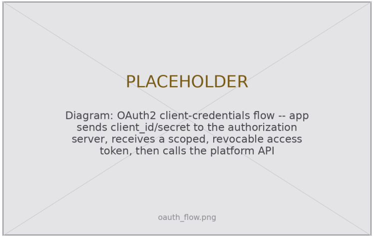
OAuth2: credentials for a scoped, revocable token

Almost every platform API authenticates with **OAuth2**. The idea: don't send your password with each request — exchange your app's credentials for a *scoped*, *revocable* token.

Two flows you should recognize:

- **Client credentials** — authenticates the *application*; grants read-only access to public data. Enough for research collection.
- **Authorization code** — additionally asks a *user* to grant your app permission to act for them. Every "Sign in with Google" dialog.

Store client IDs and secrets in environment variables, **never** in the notebook.



Bluesky app passwords and Mastodon tokens are the same delegated-credential idea, implemented more simply.

Textbook Ch. 12, "Authentication." Credential hygiene: *Missing Manual* Ch. 34.

Four platforms, four architectures

The differences are not cosmetic — they decide what data exists, on what terms, and how durable your access is:

Platform	Protocol	Auth	Rate limit	Data ownership
Reddit	Proprietary REST	OAuth2	100 QPM	Platform-controlled
Spotify	Proprietary REST	OAuth2 / client creds	∼180 RPM	Platform-controlled
Bluesky	AT Protocol *(open)*	App password	Generous, per-PDS	**User-portable**
Mastodon	ActivityPub *(open)*	OAuth2, per-instance	Varies	Instance-governed

Reddit and Spotify can *unilaterally* rewrite their terms because they own both the platform **and** the protocol. Bluesky and Mastodon cannot — their protocols are open standards anyone can implement. That is the architectural embodiment of **ownership** (Ch. 2).

Textbook Ch. 12, "Comparing Platform Architectures."

What the notebook covers

One authentication-to-DataFrame pattern, four times over:

- **Reddit (PRAW)** — subreddit metadata, submissions, comment trees, depth-vs-engagement
- **Spotify (spotipy)** — search, artist profiles, top-track popularity, user playlists
- **Bluesky (atproto)** — author feeds and the social graph (followers, following, mutuals)
- **Mastodon (Mastodon.py)** — public timelines, account search, and hashtags *across* instances

Every one: **authenticate** → **request** → **parse nested JSON** → **DataFrame.**

Connecting to Ch. 2 (Ethics): social data is people data

A person made every post, comment, and listening record. That person never expected a research dataset. Read the Terms of Service. Consider the IRB. Collect the *minimum* data that you need. Do not collect private messages. Do not quote a post if a search engine can then identify its author.

2. Wednesday · Notebook Lab

Reddit via PRAW — authenticate & explore

Client-only credentials put PRAW in read-only mode — exactly what research collection needs:

```
import praw

reddit = praw.Reddit(
    client_id="your_client_id",
    client_secret="your_client_secret",
    user_agent="WebDataScience/1.0 by u/your_username")
print(reddit.read_only) # True with app-only credentials

subreddit = reddit.subreddit("news")
print(f"Subscribers: {subreddit.subscribers:,}")
for rule in subreddit.rules: # PRAW returns Rule objects
    print("-", rule.short_name)
```

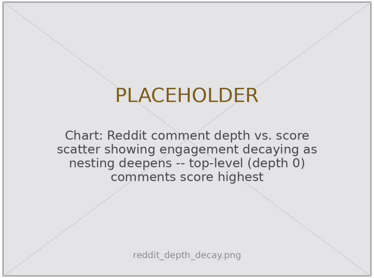
Textbook Ch. 12, “Reddit via PRAW.” Free tier: 100 queries/minute for non-commercial use.

Reddit — comment trees & the depth-decay pattern

```python
sub = next(subreddit.hot(limit=1))
sub.comments.replace_more(limit=0) # flatten tree

rows = [{"score": c.score,
         "depth": c.depth,
         "words": len(c.body.split())}
        for c in sub.comments.list()]
df = pd.DataFrame(rows)

df.groupby("depth")["score"].mean()
# decays sharply as nesting deepens
```



Depth 0 comments score highest.

Not a measure of quality — an artifact of interface design, where nested replies are progressively collapsed and harder to reach.

Textbook Ch. 12, “Analyzing Comment Engagement.” Score is skewed — prefer a Spearman rank correlation.

Spotify via spotipy — search & artist profile

The client-credentials flow reaches public catalog data — artists, albums, tracks, user playlists:

```
import spotipy
from spotipy.oauth2 import SpotifyClientCredentials

auth = SpotifyClientCredentials(client_id="...", client_secret="...")
sp = spotipy.Spotify(auth_manager=auth)

results = sp.search(q="Queen", type="artist", limit=1)
artist = results["artists"]["items"][0]
top = sp.artist_top_tracks(artist["id"])["tracks"]
print(artist["name"], f'{artist["followers"]["total"]:,}',
      [t["popularity"] for t in top[:5]]) # popularity is 0-100
```

Textbook Ch. 12, “Spotify via spotipy.” `popularity` is engagement-weighted and recomputed on Spotify’s schedule.

The audio features story — enclosure in miniature

For a decade the API's signature research offering was **audio features** — danceability, energy, valence, tempo for every track. Hundreds of studies mapped genres and modeled hits with them.

November 2024: cut off for all newly created apps. No stated research alternative — just as generative AI made music data newly valuable.

The lasting lesson: those scores were never ground truth. They were Spotify's *proprietary model* of what music *is*. You always analyze the platform's model of the world as much as the world itself.

A shrinking API

New apps get `403 Forbidden` on audio features, recommendations, and related artists. It is not a bug in your code — the access no longer exists.

Textbook Ch. 12, "The Audio Features Story." Enclosure & the post-API age (Ch. 2).

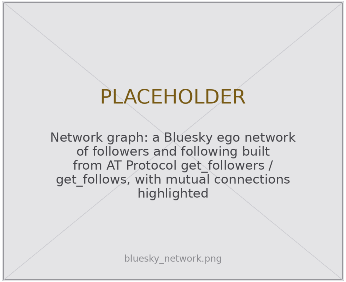
Bluesky via the AT Protocol — open by design

```python
from atproto import Client

client = Client()
client.login("you.bsky.social",
             "your-app-password")

feed = client.app.bsky.feed.get_author_feed(
    {"actor": "bsky.app", "limit": 50})

posts = [{"text": i.post.record.text,
          "likes": i.post.like_count,
          "reply": i.post.record.reply is not None}
         for i in feed.feed]
df = pd.DataFrame(posts)
```



Social graph via `get_followers` / `get_follows`.

An **app password** is a revocable delegated credential — OAuth's principle, simpler. Identity and graph are portable across any AT-Protocol service.

Textbook Ch. 12, "Bluesky via AT Protocol." No single company controls access.

Mastodon via Mastodon.py — federated, per-instance

Each instance runs its own API — there is no single endpoint for "all of Mastodon." Statuses arrive as HTML, so strip tags before analysis:

```
from mastodon import Mastodon
from bs4 import BeautifulSoup

mastodon = Mastodon(access_token="user_cred.secret",
                    api_base_url="https://mastodon.social")

for toot in mastodon.timeline_public(limit=10):
    text = BeautifulSoup(toot["content"], "html.parser").get_text()
    print("@" + toot["account"]["username"], text[:80])

tagged = mastodon.timeline_hashtag("datascience", limit=20)
```

Comprehensive collection means engaging *many* instances — each with its own rules, rate limits, and moderation norms.

Lab: work these, then show-and-tell

Register your own apps first. Then work in pairs on these exercises:

1. Compare two subreddits about one topic. Report the submissions, the engagement, and the moderation.
2. Analyze the full discography of one artist. Report the change in release frequency. Report the change in the popularity of the older albums.
3. Get a profile, a graph, and posts from Bluesky. Do the same for Reddit. Compare the difficulty and the detail.
4. Select two playlists that users made. Compare the distributions of their metadata.
5. Get 50 posts with `get_author_feed`. Calculate summary statistics. Report the limits of the AT Protocol.
6. **5617**: design a measurement plan for more than one platform. Compare a proprietary platform with a federated platform. Use Bruns (2019) and Davidson et al. (2023)

Show-and-Tell

Bring one of these to class:

- a bug that you could not solve
- data that you collected and find interesting
- a question for a research design

Platform goodwill is not infrastructure.

Twitter went from the most important data source in computational social science to inaccessible — in a year.

Build across open *and* closed, so no single shutdown ends your research.

3. Friday · Textbook Revisions

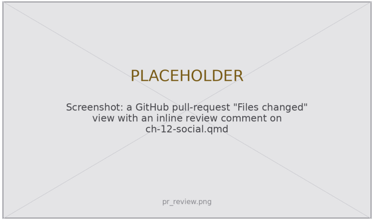
Improving Chapter 12 together

Today we make the book better. Each of you opens a **pull request** against `ch-12-social.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable

This chapter drifts *fast* — endpoints and rate limits change — so "test it against the live API" is unusually valuable here.



Contributions & reviews count toward the Textbook Revisions grade (15%).

A revision menu for Chapter 12

High-value targets — pick one and scope it to a single PR:

- **Test the code against live APIs.** Register an app, run the notebook, and report what broke: a Spotify `403`, a changed `atproto` method, a new rate limit. Fix it and note the date.
- **Close a gap in "Common Issues to Debug."** Add a worked PRAW `OAuthException` fix, a Mastodon instance-block/rate-limit case, or the `403`-diagnosis flowchart.
- **Add a figure.** An OAuth token-flow diagram, the governance-spectrum figure, or a clean comment depth-vs-score chart.
- **Strengthen a thin exercise.** The Bluesky and playlist exercises need expected output, a rubric, or starter cells so they are self-checking.
- **Refresh the comparison table & "Further Reading."** Verify mid-2026 rate limits and endpoint availability; add a Fediverse-measurement paper tied to the **ownership** callout.

Targets drawn from the chapter's callouts, "Common Issues," thin exercises, and "Further Reading."

What makes a good revision & a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **AI & Language-Model APIs** — sending prompts to models as one more data pipeline.

Key takeaways

1. One pattern, every platform: **authenticate** → **request** → **parse nested JSON** → **DataFrame**.
2. OAuth2 trades credentials for a **scoped, revocable token**; client-credentials is enough for public research reads.
3. The real variable is **governance**, not code: who owns the data, on what terms, and what happens when the rules change.
4. Open protocols (AT, ActivityPub) make access an **architectural feature**; proprietary APIs make it a revocable business decision.
5. Access is conditional, costly, and fragile — build across open **and** closed so no single closure ends your research.

Reading: Textbook Ch. 12. Related: Baumgartner et al. (2020); Diener et al. (2026); Bruns (2019).